

LLM Fine-Tuning Beginner Notes

1. What is Fine-Tuning?

Fine-tuning means taking a pre-trained language model and teaching it a specific task using your own dataset.

2. Base Model

A base model (Gemma, LLaMA, Mistral, DeepSeek) already knows language.

You do NOT train from scratch. You only adapt it.

3. Dataset Format

Most common format is JSONL or Alpaca format:

```
{  
  "instruction": "...",  
  "input": "...",  
  "output": "..."  
}
```

4. Why JSONL?

One training example per line.

Easy to stream and load.

5. Alpaca Format

Text format:

Instruction + Input + Response.

Helps models learn structured behavior.

6. LoRA (Low Rank Adaptation)

Instead of training full model, we train small adapter layers.

Faster, cheaper, needs less GPU.

7. 4-bit / 8-bit / 16-bit

Bit size affects memory, not intelligence.

4-bit is enough for most use cases.

8. Small Dataset Reality

Small data risks overfitting.

Use:

- 2 or 3 epochs
- low learning rate
- strict format

9. Training Flow

Collect data → Convert to JSONL → Format → Train → Test → Deploy

10. What Model Can Do After Training

Generate DSL from GUI steps.

Answer structured tasks.

Act as automation brain.

11. Important Mistakes to Avoid

- Too many epochs
- Bad formatting
- Inconsistent prompts

12. Real Rule

Data quality matters more than model size.